Multimedia Appendix 10. Publications that used the Unified Medical Language System in patient care.

Author	Publication year	Title	What was UMLS used for?
Miller, et al[1]	1992	CHARTLINE: providing bibliographic references relevant to patient charts using the UMLS Metathesaurus Knowledge Sources	CHARTLINE, provide bibliographic references, UMLS
Schuyler, et al[2]	1992	SIG/MED and SIG/ALP—decision-support systems in the Unified Medical Language System (UMLS)	UMLS, decision support system
Cimino[3]	1995	Use of the Unified Medical Language System in patient care at the Columbia-Presbyterian Medical Center	UMLS, patient care
Johnson, et al[4]	1997	The rubber meets the road: integrating the Unified Medical Language System Knowledge Source Server into the computer-based patient record	Integration of UMLS with patient's records, integration of codification into clinical workflow
Geissbuhler, et al[5]	1998	Clinical application of the UMLS in a computerized order entry and decision-support system	UMLS application, CPOE, CDSS
Van Mulligen[6]	1998	UMLS-based access to CPR data	UMLS, mapping CPR terms to UMLS concepts
Van Mulligen[7]	1999	UMLS-based access to CPR data. Unified Medical Language Systems	UMLS-based access to CPR, mapping CPR terms to UMLS concepts
Boxwala, et al[8]	2003	Coverage of patient safety terms in the UMLS metathesaurus	UMLS coverage, patient safety
Coonan, et al[9]	2004	Medical Informatics Standards Applicable to Emergency Department Information Systems: Making Sense of the Jumble	Medical informatics standards, UMLS as an example, emergency department
Currie, et al[10]	2004	Development and representation of a fall-injury risk assessment instrument in a clinical information system	Fall risk assessment instrument, inpatient safety, injury prevention
Kawamoto, et al[11]	2006	Design, implementation, use, and preliminary evaluation of an UMLS-enabled terminology Web service for clinical decision support	UMLS for clinical decision support, inter-vocabulary translation, terminology services
Tolentino, et al[12]	2006	Concept negation in free text components of vaccine safety reports	Negation identification, adverse events, vaccine safety

Tolentino, et al[13]	2007	A UMLS-based spell checker for natural language processing in vaccine safety	Vaccine safety, vaccine adverse events, UMLS concepts extraction
Lee, et al[14]	2008	Effectiveness of automatic acute stroke alert system based on UMLS mapped local terminology codes at emergency department	Automatic acute stroke alert system, emergency department
Pires, et al[15]	2008	A UMLS interoperable solution to support collaborative diagnosis decision making over the internet	UMLS interoperable solution, clinical decision support
Hashmi, et al[16]	2009	Computerization framework for clinical practice guidelines by extending the XML guidelines element model (GEM)	Clinical practice guidelines, computerization framework
Bodenreider, et al[17]	2010	Investigating drug classes in biomedical terminologies from the perspective of clinical decision support	Drug classes, UMLS, CDS, drug-drug interaction
Fujita, et al[18]	2010	Virtual doctor system (VDS): medical decision reasoning based on physical and mental ontologies	UMLS for testing, emotional modeling and physical views, mental ontology
Bonacin, et al[19]	2011	Careflow personalization services: concepts and tool for the evaluation of computer-interpretable guidelines	Computer-interpretable Clinical Guidelines, personalized treatment plans,
McCoy, et al[20]	2011	A prototype knowledge base and SMART app to facilitate organization of patient medications by clinical problems	Problem-medication links, medication organization
Lehman, et al[21]	2012	Risk stratification of ICU patients using topic models inferred from unstructured progress notes	Hospital mortality prediction, ICU risk stratification, UMLS
Sanchez-Garzon, et al [22]	2012	An approach for representing and managing medical exceptions in care pathways based on temporal hierarchical planning techniques	Personalized care pathways, rule represented by UMLS
Winnenburg, et al[23]	2012	Issues in creating and maintaining value sets for clinical quality measures	Clinical quality measures, value sets, UMLS
Patil, et al[24]	2013	Estimating personalized risk ranking using laboratory test and medical knowledge (UMLS)	Personalized risk ranking, lab test, UMLS
Valkenhoef, et al[25]	2013	ADDIS: A decision support system for evidence-based medicine	Safety evaluation of medical treatments, decision support system, UMLS

MacKellar, et al[26]	2014	Patient-Centered Clinical Trials Decision Support using Linked Open Data	Clinical trials query, UMLS, patient's decision support
Kuang, et al[27]	2015	Representation of Functional Status Concepts from Clinical Documents and Social Media Sources by Standard Terminologies	UMLS coverage of functional status, outcome measures
Shivade, et al[28]	2015	Comparison of UMLS terminologies to identify risk of heart disease using clinical notes	Automatic identification of risk factors for heart diseases
Varghese, et al[29]	2015	Standardized Cardiovascular Quality Assurance Forms with Multilingual Support, UMLS Coding and Medical Concept Analyses	Standardized quality assurance, cardiovascular procedures, UMLS concepts
Varghese, et al[30]	2015	Standardized quality assurance forms for organ transplantations with multilingual support, open access and UMLS coding	Clinical quality assurance, organ transplantation, UMLS for form elements
Choong, et al[31]	2017	Linking clinical quality indicators to research evidence - a case study in asthma management for children	Clinical quality indicator, outcome measures, asthma management
Moreira, et al[32]	2017	Enhancing Collaborative Case Diagnoses Through Unified Medical Language System-Based Disambiguation: A Case Study of the Zika Virus	Collaborative Case Diagnoses, Zika virus, UMLS, disambiguation
Bakal, et al[33]	2018	Exploiting semantic patterns over biomedical knowledge graphs for predicting treatment and causative relations	NLP, causal relations identification, supervised predictive models
Le, et al[34]	2018	Risk prediction using natural language processing of electronic mental health records in an inpatient forensic psychiatry setting	Risk of harm prediction, psychiatry, UMLS
Modaresnezhad, et al[35]	2019	A rule-based semantic approach for data integration, standardization and dimensionality reduction utilizing the UMLS: Application to predicting bariatric surgery outcomes	Predicting bariatric surgery outcomes, rule-based semantic networks, dimensionality reduction

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